

Module Handbook

M.Eng. International Automotive Engineering



Faculty of Electrical Engineering and Computer Science

As per: 2025-07-04

Inhalt

1	Introduction.....	3
1.1	Overview	4
1.2	Graduation	5
1.3	Degree Programme Coordination and Study Counseling	6
2	Basic Structure of the Programme	7
2.1	Compulsories	8
2.2	Electives	10
2.3	Group Projects	11
3	Description of Modules	12
3.1	Compulsory Modules.....	13
	Principles of Programming and Software Development	14
	Principles of Signal Processing, Data Analysis and Machine Learning	16
	Principles of Mathematical Modelling and Control Engineering	18
	Automotive Electrical and Electronic Technology	20
	Vehicle Dynamics	22
	Power Supply and Energy Distribution	24
	Sensor Technology and Statistical Signal Processing	26
	Automotive ECUs and Software Architectures	28
	Automotive Systems Engineering Processes	30
	Electric Drive Train	31
	Integrated Safety and Assistance Systems.....	33
	Testing and Simulation Methods for Vehicle Safety	35
	Engineering Project.....	37
	Research Methods	39
	Master Thesis	41

1 Introduction

1.1 Overview

In the field of automotive development, strong efforts should be made on national and international level to adequately prepare students for coping with the technical exigencies of future automobiles. Engineers with interdisciplinary knowledge in mechanics, electronics and computer science are particularly wanted. The Master-programme "International Automotive Engineering" (IAE) wants to impart dedicated engineering approaches for the development of automotive mechatronic systems and to give instructions for solving specific problems of developing automotive electronic systems in general as well as for vehicle safety systems specifically.

The programme takes four semesters. Semester 1 prepares students that have been graduated from different engineering disciplines for the subsequent modules. The following two semesters are dedicated to lectures, seminars and projects. The third semester is reserved for the Master's thesis. The curriculum of the Master's programme has been tailored towards the intermediation of expertise that is required to work on problems in development of electronic systems in automobiles. It mediates the special of the engineer-scientific approach. It explains the means of language and symbols to be used in automobile projects. However, scientific oriented work in a master programme means that students learn independently and solely responsible.

Multi-disciplinary modules structure the programme. The subjects of the modules emanate from mechanical engineering, electrical engineering, mathematics and engineering methodology.

Master-Thesis and Colloquium					
Automotive Systems Engineering Processes	Electric Drive Train	Sensor Technology and Statistical Signal Processing	Power Supply and Energy Distribution	Scientific Elective	Project
Vehicle Dynamics	Testing and Simulation Methods for Vehicle Safety	Integrated Safety and Assistance Systems	Automotive ECU and Software Architectures	Scientific Elective	German Language Elective
Principles of Programming and Software Development (6 SWS / 7 ECTS)	Principles of Signal Processing, Data Analysis and Machine Learning (6 SWS / 7 ECTS)	Principles of Mathematical Modelling and Control Engineering (4 SWS / 6 ECTS)	Electrical and Electronic Technology (4 SWS / 5 ECTS)	Project & Scientific Seminar	

Figure 1: General Programme structure

Compulsory modules aim at transfer of knowledge an automotive engineer must have. The compulsory module Project enables students to incorporate into a new complex task and - based on a division of labor - to work on this task interdisciplinary in a team using suitable scientific methods.

1.2 Graduation

The Technische Hochschule Ingolstadt awards the academic degree

Master of Engineering (M.Eng.)

1.3 Degree Programme Coordination and Study Counseling

For subject-oriented questions and problems, the course advisor is available:

Prof. Dr. Armin Arnold

Questions related to the organization will be answered by:

Prof. Dr. Armin Arnold

The consultation hours that apply during the semester are announced via Moodle.

2 Basic Structure of the Programme

The Master's programme starts every summer and winter semester. Due to the modular structure of the degree programme it is possible to complete all subjects both at the beginning in the summer and at the beginning in the winter semester. Therefore, not every subject is offered every semester. The following two tables represent the curriculum for a study start in the winter semester or in the summer semester.

2.1 Compulsories

Start in winter

SPO-No.	Module	Semester 1		Semester 2		Semester 3	
		SWS/ECTS	Exam	SWS/ECTS	Exam	SWS/ECTS	Exam
1	Principles of Programming and Software Development	6 / 7	PF				
2	Principles of Signal Processing, Data Analysis and Machine Learning	6 / 7	PF				
3	Principles of Mathematical Modelling and Control Engineering	4 / 6	schrP				
4	Automotive Electrical and Electronic Technology	4 / 5	schrP				
14	Engineering Project	2 / 2.5	Proj				
15	Research Methods	2 / 2.5	SA				
6	Vehicle Dynamics			4 / 5	schrP		
9	Automotive ECU and Software Architectures			4 / 5	PF		
12	Integrated Safety and Assistance Systems			4 / 5	schrP		
13	Testing and Simulation Methods for Vehicle Safety			4 / 5	schrP		
17	Scientific Elective			4 / 5	LN		
5	German Language Elective			4 / 5	LN		
7	Power Supply and Energy Distribution					4 / 5	schrP
8	Sensor Technology and Statistical Signal Processing					4 / 5	schrP
10	Automotive Systems Engineering Processes					4 / 5	mdIP
11	Electric Drive Train					4 / 5	schrP
16	Team Project					2 / 5	Proj
17	Scientific Elective					4 / 5	LN
Sum		24/30		24/30		22/30	

schrP written exam

Proj project work

SA seminar paper

mdIP oral exam

LN subject-defined exam

PF portfolio exam

Start in summer

SPO-No.	Module	Semester 1		Semester 2		Semester 3	
		SWS/ECTS	Exam	SWS/ECTS	Exam	SWS/ECTS	Exam
1	Principles of Programming and Software Development	6 / 7	PF				
2	Principles of Signal Processing, Data Analysis and Machine Learning	6 / 7	PF				
3	Principles of Mathematical Modelling and Control Engineering	4 / 6	schrP				
4	Automotive Electrical and Electronic Technology	4 / 5	schrP				
14	Engineering Project	2 / 2.5	Proj				
15	Research Methods	2 / 2.5	SA				
7	Power Supply and Energy Distribution			4 / 5	schrP		
8	Sensor Technology and Statistical Signal Processing			4 / 5	schrP		
10	Automotive Systems Engineering Processes			4 / 5	mdIP		
11	Electric Drive Train			4 / 5	schrP		
17	Scientific Elective			4 / 5	LN		
5	German Language Elective			4 / 5	LN		
6	Vehicle Dynamics					4 / 5	schrP
9	Automotive ECU and Software Architectures					4 / 5	PF
12	Integrated Safety and Assistance Systems					4 / 5	schrP
13	Testing and Simulation Methods for Vehicle Safety					4 / 5	schrP
16	Team Project					4 / 5	LN
17	Scientific Elective					4 / 5	LN
Sum		24/30		24/30		22/30	

schrP written exam

Proj project work

SA seminar paper

mdIP oral exam

LN subject-defined exam

PF portfolio exam

2.2 Electives

Required elective modules are modules offered to students of the degree programme. Each student must complete a total of two elective modules according to the study and examination regulations. The selected modules are treated like compulsory modules. A claim that all envisaged elective modules are actually offered does not exist. Likewise, there is no claim that the associated teaching events are carried out if the number of participants is insufficient. Which modules are offered in the respective semester can be found in the curriculum.

2.3 Group Projects

In group projects, a semester-accompanying project task is done by a team of about 10-12 students.

Selecting a group projects is as follows:

In the week before the beginning of the semester, students are asked online to choose the project they are interested in. Due to the limited number of participants per project, it cannot be guaranteed that each student will get a place in his preferred project. Students are encouraged to independently organize project changes.

Before the selection of the projects take place students will be informed about the topics and tasks of the projects offered in the semester.

As part of the examination registration, students have to register which project they should complete with which lecturer.

There is no claim that all planned projects will be actually offered.

3 Description of Modules

3.1 Compulsory Modules

Principles of Programming and Software Development			
Module abbreviation:	IAE_ProgSW	Reg.no.:	1
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Hagerer, Andreas		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	7 ECTS / 6 SWS		
Workload:	Contact hours:	70 h	
	Self-study:	105 h	
	Total:	175 h	
Subjects of the module:	1: Principles of Programming and Software Development		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
PF - Portfolio Exam			
Additional Explanation:			
The portfolio exam aims to assess your ability to apply fundamental programming and software development skills in real-world and problem-solving contexts. It consists of three individual examinations			
• a programming assignment to be submitted online (10%) • a programming project to be submitted online with an interview about code produced (20%) • a computer based written exam (90 minutes, closed book) asking with programming and software development concepts comprehension and writing of short code pieces (70%)			
If an individual examination is not attended, the overall grade will be “not attended” (in German: nicht teilgenommen).			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
Upon successful completion of this course students will be able to:			
• use lists, tuples, and dictionaries in Python programs, • to apply control structures like loops and decision statements, • design and implement programs to solve real-world problems in Python, • understand the basic structure of NumPy arrays, • create, manipulate and use Numpy arrays efficiently to solve linear algebra problems, • be familiar with software development concepts like software lifecycle, software specification and process models, software project management, and quality standards, • apply methods used to plan, develop and engineer software solutions			

Content:
• information representation and manipulation by means of variables, data structures, statements and control structures • structuring programs by means of functions and modules • introduction to object oriented programming • modules and concepts of the scientific Python ecosystem: NumPy, Matplotlib, Scikit-Learn • phases and characteristics of software processes • concepts of requirements engineering, software modeling, design and implementation
Literature:
• MATTHES, Eric, 2023. <i>Python crash course: a hands-on, project-based introduction to programming</i>. San Francisco: No Starch Press. ISBN 978-1-7185-0270-3 • GUTTAG, John, 2021. <i>Introduction to computation and programming using Python: with application to computational modeling and understanding data</i>. Cambridge, MA: The MIT Press. ISBN 978-0-262-36343-3 • PINE, David J., 2019. <i>Introduction to Python for science and engineering</i> [online]. Boca Raton ; London ; New York: CRC Press PDF e-Book. ISBN 978-0-429-01425-3, 978-0-429-50641-3. Available via: https://doi.org/10.1201/9780429506413. • LANGTANGEN, Hans Petter, 2016. <i>A primer on scientific programming with Python</i> [online]. Berlin: Springer PDF e-Book. ISBN 978-3-662-49887-3. Available via: https://doi.org/10.1007/978-3-662-49887-3.
Additional remarks:
None

Principles of Signal Processing, Data Analysis and Machine Learning			
Module abbreviation:	IAE_SigML	Reg.no.:	2
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Mecking, Michael		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	7 ECTS / 6 SWS		
Workload:	Contact hours:	70 h	
	Self-study:	105 h	
	Total:	175 h	
Subjects of the module:	2: Principles of Signal Processing, Data Analysis and Machine Learning		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	This module is not compulsory in any other degree programme within the faculty. When transferring to a different faculty, credit transfer must be discussed with the responsible person.		
Examinations:			
PF - Portfolio Exam			
Additional Explanation:			
The examination will consist of two individual exams:			
- A 45-minute written exam before the examination period (33% overall weight), - A 90-minute written exam during the examination period (67% overall weight).			
If an individual examination is not attended, the overall grade will be “not attended” (in German: nicht teilgenommen).			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
A thorough understanding of complex variables and mathematical analysis is required.			
Objectives:			
Upon completion of the module, students will be able to			
- describe and analyse the properties of signals and discrete-time linear systems in the time and frequency domain using discrete-time transforms, - design linear discrete-time filters with prescribed properties, - determine the limitations and impact of sampling and quantisation on the conversion of analogue to digital signals, - apply (fast) Fourier-Transforms to analyse and realise discrete-time signal processing systems, - understand statistical machine learning (ML), - identify ML potential and limitations in application domains, - choose appropriate ML methods, models, and algorithms for a given problem, - analyze properties of ML models and algorithms.			

Content:
1. Fundamentals of Discrete-Time Signals and Systems 2. System Analysis and Transform Techniques 3. Sampling Theory 4. Digital Filter Design and Filter Implementation 5. Classification tasks and basic learning algorithms 6. Model Architectures and Learning Techniques 7. Models for Sequential Data 8. Unsupervised Learning
Literature:
• OPPENHEIM, Alan V. and Ronald W. SCHAFER, 2014. <i>Discrete-time signal processing</i>. Harlow: Pearson. ISBN 978-1-292-03815-5 • VAIDYANATHAN, P. P., 2024. <i>Signals, systems, and signal processing</i>. Cambridge: Cambridge University Press. ISBN 978-1-00-941229-2 • MURPHY, Kevin P., 2022. <i>Probabilistic machine learning: an introduction</i>. Cambridge, Massachusetts: The MIT Press. ISBN 978-0-262-36930-5 • VAN DER PLAS, Jake. <i>Python Data Science Handbook</i> [online]. PDF e-Book. Available via: https://jakevdp.github.io/PythonDataScienceHandbook/.
Additional remarks:
None

Principles of Mathematical Modelling and Control Engineering			
Module abbreviation:	IAE_MathCtrlEng	Reg.no.:	3
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Gregor, Rudolf		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	6 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	103 h	
	Total:	150 h	
Subjects of the module:	3: Principles of Mathematical Modelling and Control Engineering		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
After a successful participation of this course students are able to - investigate real technical systems, to structure them into appropriate subsystems and derive suitable differential equations or systems of differential equations, - analyze control systems and create models in the time and frequency domain, - analyze systems with regards to dynamics, oscillatory behavior and stability, - understand the system specifications which a closed-loop circuit must fulfill, - evaluate and select suitable controller structures in accordance to the specifications, - apply different methods to parameterize the controller structures and analyze the results in simulations.			
Content:			
- Methodological approaches to derive ordinary differential equations from simple electrical and mechanical systems - Different methods to implement linear and non-linear differential equations in Matlab/Simulink (transfer functions, state space representation, ...) - Laplace-Transformation and its fields of application (implementation and analysis of LTI-systems) - System description and representation in the time and frequency domain			

• Elementary control loop elements • Control loops: requirements, behavior, design • Controller synthesis: root locus / Bode diagram / empirical approach
Literature:
• SUN, Jing, 2018. <i>Control engineering: fundamentals</i> [online]. Berlin: De Gruyter PDF e-Book. ISBN 978-3-11-057327-5. Available via: https://doi.org/10.1515/9783110573275. • BOLTON, William, 2010. <i>Control engineering</i>. Harlow u.a.: Prentice Hall. ISBN 978-0-582-32773-3 • DORF, Richard C. and Robert H. BISHOP, 2022. <i>Modern control systems</i>. Harlow, United Kingdom: Pearson. ISBN 978-1-292-42235-0
Additional remarks:
None

Automotive Electrical and Electronic Technology			
Module abbreviation:	IAE_ElecTechn	Reg.no.:	4
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Arnold, Armin		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	4: Automotive Electrical and Electronic Technology		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
After successfully completing the module, the students have a • knowledge of automotive electronics basics. • knowledge of the electric architecture of automotive control units and applied integrated circuits. • knowledge of automotive sensor and actuator technologies. • knowledge of automotive bus systems.			
Content:			
• basics of electrical and electronic engineering • introduction to sensor and actuator technology • basic functionalities of different bus systems • control unit circuits for input and sensor signal conditioning, output drivers and controlling actuators, power supply • physical layer of automotive communication networks and onboard communication • introduction to automotive electric standards			

Literature:
• ZAMAN, Najamuz, 2015. <i>Automotive electronics design fundamentals</i> [online]. Cham [u.a.]: Springer PDF e-Book. ISBN 978-3-319-17584-3. Available via: https://doi.org/10.1007/978-3-319-17584-3. • IDA, Nathan, 2021. <i>Engineering Electromagnetics</i> [online]. Cham: Springer International Publishing PDF e-Book. ISBN 978-3-030-15557-5. Available via: https://doi.org/10.1007/978-3-030-15557-5. • ROBERT BOSCH GMBH (ED.), 2014. <i>Bosch Automotive Electrics and Automotive Electronics: Systems and Components, Networking and Hybrid Drive</i> [online]. PDF e-Book. ISBN 978-3-658-01784-2. Available via: https://doi.org/10.1007/978-3-658-01784-2.
Additional remarks:
None

Vehicle Dynamics			
Module abbreviation:	IAE_VDS	Reg.no.:	6
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Arnold, Armin		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	6: Vehicle Dynamics		
Lecture types:	1: SU/Ü - seminaristischer Unterricht/Übung		
Availability of the module:	This module is not a compulsory module in any other degree programme of the faculty. When changing to a degree programme of another faculty, the possibilities for credit transfer must be discussed with the responsible persons.		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
ability to apply the physical rules of mechanical systems, especially Newton's laws; basic knowledge of electrics/electronics			
Objectives:			
After successfully completing the module the students shall be able to			
• explain and judge all tire properties that are important for vehicle dynamics • calculate according to some simplified vehicle models • analyse how drivetrain, brakes and other chassis components work together, e.g. like control arms, spring rates, position of center of gravity, differentials including limited slip differentials, torque-vectoring-differentials • explain ABS-control • explain vehicle stability control systems • deduct the additional possibilities given by four-wheel-steering, torque-vectoring and active suspensions			
Content:			
• Tire and tire properties under different conditions (camber, normal force, combinations of longitudinal and/or lateral slip • Kamm's circle and its application to different scenarios • Properties of rubber			

• Brush model of tire • Vehicle models (Single track model steady state and dynamically, application to cornering, banked road,, sidewind, iphysical and effective sideslip stiffness) • Influencing driving behaviour by: means of suspension: ○ Suspension:: Roll- und instant center, (elasto)-kinematics ○ Spring stiffnesses ○ position of center of gravity ○ Distribution of driving- and braking torques • Drive train influences: Behavior of open differentials, limited slip differentials (viscous and lclutch type, 4WD) • ABS algorithm • traction control and vehicle stability control • torque vectoring
Literature:
• REIMPELL, Jörnßen, Jürgen W. BETZLER and Helmut STOLL, 2001. <i>The automotive chassis: engineering principles: chassis and vehicle overall, wheel suspensions and types of drive, axle kinematics and elastokinematics, steering - springing - tyres, construction and calculations advice</i>. Oxford [u.a.]: Butterworth-Heinemann. ISBN 0-7506-5054-0 • MILLIKEN, William F. and Douglas L. MILLIKEN, 1995. <i>Race car vehicle dynamics</i>. Warrendale, PA: SAE International. ISBN 1-56091-526-9, 978-1-56091-526-3 • GENTA, Giancarlo and Lorenzo MORELLO, . <i>The automotive chassis</i>. [Dordrecht]: Springer Netherland. • HANEY, Paul, 2012. <i>The racing & high-performance tire: using the tires to tune for grip and balance</i>. Dallas, Tex. [u.a.]: InfoTire [u.a.]. ISBN 0-9646414-2-9, 978-0-7680-12415
Additional remarks:
None

Power Supply and Energy Distribution			
Module abbreviation:	IAE_PSED	Reg.no.:	7
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Pforr, Johannes		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	7: Power Supply and Energy Distribution		
Lecture types:	SU/Ü - seminaristischer Unterricht/Übung		
Availability of the module:	This module is not a compulsory module in any other degree programme of the faculty. When changing to a degree programme of another faculty, the possibilities for credit transfer must be discussed with the responsible persons.		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
Basic knowledge of electronics			
Objectives:			
After successfully completing the module the students should			
• have good knowledge in the field of modern energy distribution systems in cars and of the components used in the automotive energy nets • understand why energy management systems are important for the operation of electric energy nets in cars • understand the operation principle of power electronic converters for automotive applications • understand and to use methods to develop steady-state and dynamic models of power electronic converters for given type of problems • analyze and judge the steady-state and dynamic performance of automotive electrical energy nets with power electronic components according to given targets • understand the operation principle of modern electric machines for electric and hybrid electric vehicles including the control of the electric machines • be able to use steady-state and dynamic models of electric machines in order to analyze the energy flow in automobile electrical energy nets dependent on the operation strategy of the vehicle • be able to derive models of given automotive energy nets and the components and to perform simulations for optimization purposes			

Content:

Introduction, background and design of vehicular electrical energy distribution networks and power electronic systems and devices:

- Power Devices and Converter Topologies
- 14V / 48V Power Supply and Energy Distribution
- Generation of electric Power in Vehicles
- Energy management Systems
- High Voltage electric Energy Distribution for Hybrid Vehicles
- Electric motor Drives and motion Control
- Starter / Generator
- Simulation

Literature:

- VELTMAN, André, PULLE, Duco W. J., DE DONCKER, Rik W., 2016. *Fundamentals of Electrical Drives* [online]. Cham: Springer International Publishing PDF e-Book. ISBN 978-3-319-29409-4, 978-3-319-29408-7. Available via: <https://doi.org/10.1007/978-3-319-29409-4>.
- ERICKSON, Robert W. and Dragan MAKSIMOVIĆ, 2004. *Fundamentals of power electronics*. Dordrecht: Kluwer. ISBN 0-7923-7270-0, 978-0-7923-7270-7
- LEONHARD, Werner, 2001. *Control of electrical drives*. Berlin [u.a.]: Springer. ISBN 3-540-41820-2
- EHSANI, Mehrdad, Yimin GAO and Ali EMADI, 2010. *Modern electric, hybrid electric, and fuel cell vehicles: fundamentals, theory, and design*. Boca Raton, FL [u.a.]: CRC Press, Taylor & Francis Group. ISBN 978-1-4200-5400-2, 978-1-4200-5398-2

Additional remarks:

No remarks.

Sensor Technology and Statistical Signal Processing			
Module abbreviation:	IAE_ST&SP	Reg.no.:	8
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	General Elective Subject	1
Responsible for module:	Botsch, Michael		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	8: Sensor Technology and Signal Processing		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
After successfully completing the module the students are able to • describe major trends in the automotive sensor market; • categorize automotive sensors with respect to the underlying physical effects; • to analyze sensor signals in the time- and frequency-domain; • apply statistical signal processing algorithms (e. g., Kalman filter) to automotive sensor data; • to evaluate algorithms for sensor data fusion; • to design and apply simple machine learning algorithms • to implement statistical signal processing algorithms in Matlab.			
Content:			
• Introduction to Automotive Sensors ○ Automotive Sensor Market ○ Sensor Technologies ○ Sensor Types and Characteristics ○ Multi-Modal Sensor Systems • Statistical Signal Processing ○ Signal Types and Characteristics			

○ Basics of Statistical Signal Processing ○ Pattern Recognition ○ Kalman Filter ● Sensor Data Fusion ○ Data Association ○ Track-To-Track Fusion ● Analog and Digital Processing of Signals ○ Analog Filters, Amplifiers and A/D Converters ○ Fourier Series and Transform, Laplace- and z-Transform ○ Digital Filters
Literature:
● , . Current publications from IEEE Symposium on Intelligent Vehicle and from IEEE International Conference on Intelligent Transportation Systems. In: . ● BOTSCH, Michael and Wolfgang UTSCHICK, 2020. <i>Fahrzeugsicherheit und automatisiertes Fahren: Methoden der Signalverarbeitung und des maschinellen Lernens</i> . ISBN 978-3-446-45326-5 ● WINNER, Hermann and others, 2024. <i>Handbuch Assistiertes und Automatisiertes Fahren: Grundlagen, Komponenten und Systeme für assistiertes und automatisiertes Fahren</i>. ISBN 3658384859 ● BAR-SHALOM, Yaakov, LI, Xiao-Rong, KIRUBARAJAN, Thiagalingam, 2001. <i>Estimation with applications to tracking and navigation</i> [online]. New York: Wiley PDF e-Book. ISBN 0-471-46521-6, 978-0-471-46521-8. Available via: http://onlinelibrary.wiley.com/book/10.1002/0471221279. ● REIF, Konrad, 2016. <i>Sensoren im Kraftfahrzeug</i> [online]. Wiesbaden: Springer Vieweg PDF e-Book. ISBN 978-3-658-11211-0, 978-3-658-11210-3. Available via: https://doi.org/10.1007/978-3-658-11211-0.
Additional remarks:
None

Automotive ECUs and Software Architectures			
Module abbreviation:	IAE_EcuArch	Reg.no.:	9
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Margull, Ulrich		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	9: Automotive ECUs and Software Architectures		
Lecture types:	SU/Ü/PR - seminar based teaching/Exercise course/laboratory		
Availability of the module:	None		
Examinations:			
PF - Portfolio Exam			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
Upon successful completion, students will be able to:			
• explain the main components of vehicle electronic/electrical systems, • recognize key operational blocks within a typical automotive ECU, • operating principles of ECU microcontrollers/microprocessors and peripherals, • identify software/firmware control blocks and the interaction with hardware • develop and design software for embedded, automotive, real-time systems using AUTOSAR.			
Content:			
• ECU fundamentals and hardware architecture • Automotive microcontroller/ microprocessors • Embedded software and fail-safe software implementation • Architecture of automotive software: modularity, software layers, real-time systems (tasks, scheduling), resource management (deadlocks, semaphores, priority inversion), interrupts and timers • Software processes: V-model and MISRA development guideline, process assessment (CMMI, automotive SPICE), model-based development (Matlab/Simulink/Stateflow), supporting processes (e.g. versioning with git, requirements tracing)			

AUTOSAR development process for Classic (Virtual Function Bus, Application Components, RTE, BSW, AUTOSAR OS) and Adaptive AUTOSAR
Literature:
- MARWEDEL, Peter, 2021. <i>Embedded system design: embedded systems foundations of cyber-physical systems, and the internet of things</i>. Cham, Switzerland: Springer. ISBN 978-3-030-60909-2, 978-3-030-60912-2 - LEE, Edward A. and Sanjit Arunkumar SESHIA, 2019. <i>Introduction to embedded systems: a cyber-physical systems approach</i>. Dehli: PHI Learning. ISBN 978-93-88028-40-0 - SCHÄUFFELE, Jörg and Thomas ZURAWKA, 2016. <i>Automotive software engineering: principles, processes, methods, and tools</i>. Warrendale, Pennsylvania, USA: SAE International. ISBN 978-0-7680-7992-0 - PANDIAN, Sivakumar, KUMAR, B. Vinoth, DEVI, R. S. Sandhya, 2022. <i>Software engineering for automotive systems: principles and applications</i> [online]. Boca Raton ; London ; New York: CRC Press PDF e-Book. ISBN 978-1-003-26990-8, 9781003269908. Available via: https://doi.org/10.1201/9781003269908. - BLOKDYK, Gerardus, 2021. <i>AUTOSAR: a complete guide ; practical tools for self-assessment</i>. 2020. edition. [Erscheinungsort nicht ermittelbar]: 5starcooks. ISBN 978-0-6559-1480-8
Additional remarks:
None

Automotive Systems Engineering Processes			
Module abbreviation:	IAE_SysEng	Reg.no.:	10
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Giesler, Björn		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	10: Automotive Systems Engineering Processes		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
mdIP - oral exam, 20 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
To be determined			
Content:			
To be determined			
Literature:			
Will be specified at the beginning			
Additional remarks:			
None			

Electric Drive Train			
Module abbreviation:	IAE_ElectDT	Reg.no.:	11
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Schiele, Thomas		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	11: Electric Drive Train		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
none			
Recommended prerequisites:			
basic knowledge of physics (Work, Power, Forces, Torques, ...), engineering mathematics (differential and integral calculus), electrical engineering fundamentals, engineering mechanics			
Objectives:			
After successfully completing the module the students			
• know details about longitudinal dynamics of vehicles and its mathematical description • show detailed knowledge of electric and hybrid drivetrain components (e-Motor, power-electronics, energy-storage systems, basics of IC engines) • understand advantages and disadvantages of different drivetrain concepts according to driving performance and energy consumption • know basic principles of operational strategies and energy management in electric and hybrid vehicles • are able to explain the principles of different charging systems • know details about vehicle storage systems and their advantages and disadvantages • are able to set up models and evaluate results from dynamic drivetrain simulations focussing on the impact of operation principles on factors like driving performance and efficiency			
Content:			
• basics of vehicle movement, driving resistances and energy consumption • electric machines for electric and hybrid vehicles • power-electronics for electric motors			

• drivetrain concepts for electric vehicles • basics of IC engines and potentials of hybridization • hybrid drivetrain concepts and operational strategies • energy storage systems for vehicle applications • charging systems for electric vehicles • fuel cell vehicles • modelling and simulation of different drivetrain concepts
Literature:
• PATEL, Nil, BHOI, Akash Kumar, PADMANABAN, Sanjeevikumar, HOLM-NIELSEN, Jens Bo, 2021. <i>Electric vehicles: modern technologies and trends</i> [online]. Singapore: Springer PDF e-Book. ISBN 978-981-159-251-5. Available via: https://doi.org/10.1007/978-981-15-9251-5. • ELGOWAINY, Amgad, 2021. <i>Electric, Hybrid, and Fuel Cell Vehicles</i> [online]. New York, NY: Springer New York PDF e-Book. ISBN 978-1-07-161492-1. Available via: https://doi.org/10.1007/978-1-0716-1492-1. • DOPPELBAUER, Martin, 2025. <i>Grundlagen der Elektromobilität: Technik, Praxis, Energie und Umwelt</i> [online]. Wiesbaden: Springer Fachmedien Wiesbaden PDF e-Book. ISBN 978-3-658-44828-8. Available via: https://doi.org/10.1007/978-3-658-44828-8. • KAMPKER, Achim, HEIMES, Heiner Hans, 2024. <i>Elektromobilität: Grundlagen einer Fortschrittstechnologie</i> [online]. Berlin, Germany: Springer Vieweg PDF e-Book. ISBN 978-3-662-65812-3. Available via: https://doi.org/10.1007/978-3-662-65812-3.
Additional remarks:
None

Integrated Safety and Assistance Systems			
Module abbreviation:	IAE_ISAS	Reg.no.:	12
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	General Elective Subject	1
Responsible for module:	Botsch, Michael		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	12: Integrated Safety and Assistance Systems		
Lecture types:	1: SU/Ü - seminaristischer Unterricht/Übung		
Availability of the module:	This module is not a compulsory module in any other degree programme of the faculty, but it is offered as an elective module in other Master degree programmes of the faculty.. When changing to a degree programme of another faculty, the possibilities for credit transfer must be discussed with the responsible persons.		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
Mathematics for Engineers			
Recommended prerequisites:			
None			
Objectives:			
After successfully completing the module the students are able - to explain basic vehicle components that are required for driver assistance systems and for vehicle integrated safety functions - to analyze and evaluate state of the art driver assistance systems - to describe testing procedures that are used for vehicle active safety functions - to explain mathematically the concepts for motion planning that are used in algorithms for driver assistance systems and integrated safety functions - to implement basic trajectory planning algorithms in Matlab.			
Content:			
- Introduction to IS & DAS - Examples of Driver Assistance and Integrated Vehicle Safety Systems: Parking Systems, Adaptive Cruise Control, Autonomous Emergency Braking - Position and Orientation: Pose, Representing Pose in 2-D and in 3-D - Time and Motion: Generation of Trajectories, Rate of Change and Inverse Problem			

• Vehicle Motion Models: Decoupled X- and Y-Dynamics, Constant Velocity Model, Constant Steering Angle and Velocity Model, Constant Turn Rate and Acceleration Model, One-Track Model, Two-Track Model • Navigation and Localization
Literature:
• KELLY, Alonzo, 2013. <i>Mobile robotics: mathematics, models, and methods</i>. New York, NY: Cambridge Univ. Press. ISBN 978-1-107-03115-9 • HEISSING, Bernd, 2011. <i>Chassis handbook: fundamentals, driving dynamics, components, mechatronics, perspectives</i> [online]. Wiesbaden: Vieweg+Teubner PDF e-Book. ISBN 978-3-8348-9789-3. Available via: https://doi.org/10.1007/978-3-8348-9789-3. • WINNER, Hermann, HAKULI, Stephan, LOTZ, Felix, SINGER, Christina, 2019-. <i>Handbook of Driver Assistance Systems: Basic Information, Components and Systems for Active Safety and Comfort</i> [online]. Cham: Springer International Publishing PDF e-Book. ISBN 978-3-319-09840-1. Available via: https://doi.org/10.1007/978-3-319-09840-1. • BOTSCH, Michael, UTSCHICK, Wolfgang, 2020. <i>Fahrzeugsicherheit und automatisiertes Fahren: Methoden der Signalverarbeitung und des maschinellen Lernens</i> [online]. PDF e-Book. ISBN 978-3-446-46804-7.
Additional remarks:
No remarks.

Testing and Simulation Methods for Vehicle Safety			
Module abbreviation:	IAE_TSMS	Reg.no.:	13
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Vaculin, Ondrej		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	5 ECTS / 4 SWS		
Workload:	Contact hours:	47 h	
	Self-study:	78 h	
	Total:	125 h	
Subjects of the module:	13: Testing and Simulation Methods for Vehicle Safety		
Lecture types:	SU/Ü - lecture with integrated exercises		
Availability of the module:	None		
Examinations:			
schrP90 - written exam, 90 minutes			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
After successfully completing the module the students			
• shall know how to test automotive safety systems and control units while its development process • shall understand different testing methods and their usage for different types of control units and different criticalities. • shall know when and how to use simulation as an improvement of the testing process, which types of simulation can be used and their pros and cons.			
Content:			
• Vehicle Approval Process, Consumer Testing • Testing as part of the development process (ISO 26262/ V-Model) • Testing methods and testing metrics ○ Electrical Safety ○ Passive Safety ○ Active Safety ○ Automated Driving • Application of simulation based methods • Components of simulation			

• Different model types
Literature:
• ARJUNRAJ, P., 2024. <i>Functional Safety in Modern Mobility: ISO 26262 and Beyond: ISO 26262 and Beyond</i>. ISBN 979-8894753379 • THE ART SERVICE, 2020. <i>ISO 26262 A Complete Guide</i>. ISBN 978-1867441762 • SOKOLOVSKIJ, Edgar, ŽURAULIS, Vidas, 2024. <i>Advances in vehicle dynamics and road safety: technologies, simulations and applications</i> [online]. Basel: MDPI PDF e-Book. ISBN 978-3-7258-1182-3. Available via: https://doi.org/10.3390/books978-3-7258-1182-3.
Additional remarks:
None

Engineering Project			
Module abbreviation:	IAE_EngProj	Reg.no.:	14
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Einsetzungstext ist leer!	1
Responsible for module:	Arnold, Armin		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	2.5 ECTS / 2 SWS		
Workload:	Contact hours:	23 h	
	Self-study:	40 h	
	Total:	63 h	
Subjects of the module:	14: Engineering Project		
Lecture types:	Prj - project		
Availability of the module:	None		
Examinations:			
Project report and oral presentation 15 min.			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
Upon completing the module, students - will have gained a deeper understanding of the complexity and challenges of engineering project work in a small team, - will have an understanding of the benefits of soft skills such as communication, moderation, forward-looking project planning and team-based task distribution - will have applied engineering knowledge to a project and learned methods to enable interdisciplinary collaboration and to organize cooperation at interfaces and smooth workflows across disciplines - will know the necessity of working to deadlines and have practiced meeting milestone targets in an engineering work environment.			
Content:			
The module combines the practical application of engineering knowledge and methods with interdisciplinary key skills such as project management and team skills, in particular communication and conflict resolution skills.			
Groups of up to 8 students are formed, who work during the semester on a project offered by the supervisors and participating chairs. Through a combination of theoretical training and supervised team/project work, in which students make decisions largely independently, they learn to work interactively across discip-			

lines as part of a team. By jointly planning the approach to the project and dealing with challenging implementation problems, students gain knowledge about project work and the necessary key qualifications in addition to specialist skills.
Literature:
Will be specified at the beginning
Additional remarks:
None

Research Methods			
Module abbreviation:	IAE_ResSem	Reg.no.:	15
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	1
Responsible for module:	Vaculin, Ondrej		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	2.5 ECTS / 2 SWS		
Workload:	Contact hours:		23 h
	Self-study:		40 h
	Total:		63 h
Subjects of the module:	15: Research Methods		
Lecture types:	S - seminar		
Availability of the module:	None		
Examinations:			
SA - Seminar paper with oral examination (15min) and written elaboration (8-15 pages)			
Additional Explanation:			
None			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			
Objectives:			
Upon completion of the module, students will be able to			
• acquire knowledge in a technical field by means of independent literature research and analysis. • summarise and explain a technical content in the form of a brief written elaboration following scientific referencing guidelines. • present this knowledge comprehensibly within the framework of an oral presentation using suitable supportive media. • understand a presentation and to critically and professionally discuss the contents with the speaker. • apply enhanced interdisciplinary knowledge and communicative skills in relevant contexts.			
Content:			
• The respective lecturer compiles a collection of publications from the technical literature. • In the course of the seminar, each student is required to present a paper/topic that was assigned by either lot or choice at the beginning of the semester. • In the preparation phase, each student must conduct an independent literature research on the topic and aggregate the results. • The student will give an oral presentation on the topic lasting about 25 minutes followed by a discussion with peers. Participation in the discussions is contributing to the final grade.			

- In addition to an oral presentation, the student is required to prepare a written paper on the topic of the presentation. This paper should summarise the main contents.

The respective instructor will communicate detailed information on topics, deadlines and expectations regarding the presentation as well as the written elaboration at the beginning of the semester.

Literature:

Will be specified at the beginning

Additional remarks:

None

Master Thesis			
Module abbreviation:	IAE_THESIS	Reg.no.:	18
Curriculum:	Programme	Module type	Semester
	International Automotive Engineering (SPO WS 25/26)	Compulsory Subject	
Responsible for module:	Arnold, Armin		
Language of instruction:	English	Language of exam:	English
Credit points / SWS:	30 ECTS / 0 SWS		
Workload:	Contact hours:		0 h
	Self-study:		750 h
	Total:		750 h
Subjects of the module:	18: Master's Thesis 18.1: Master's Thesis 18.2: Seminar for Master's thesis		
Lecture types:	Master Thesis: unspecified Master Thesis: unspecified Master Thesis Colloquium: S - seminar		
Availability of the module:	None		
Examinations:			
Master Thesis: Master-Thesis Master Thesis: Master-Thesis Master Thesis Colloquium: Additional Explanation: In general, students look for a topic for their thesis on their own. Topics are either offered internally by university lecturers in notices (also online) or result from the cooperation of the student with a company. In the case of an externally provided topic, the student must convince a university lecturer of his or her topic so that the lecturer assumes the role of the first examiner. For this purpose, it is advisable to outline the topic and the planned approach in a short paper. This exposé serves to convince the lecturer desired as the first examiner. In general, students look for a topic for their thesis on their own. Topics are either offered internally by university lecturers in notices (also online) or result from the cooperation of the student with a company. In the case of an externally provided topic, the student must convince a university lecturer of his or her topic so that the lecturer assumes the role of the first examiner. For this purpose, it is advisable to outline the topic and the planned approach in a short paper. This exposé serves to convince the lecturer desired as the first examiner. In the course of the master thesis seminar, students must participate in regular meetings with the supervising professor/lecturer (first examiner).			
Prerequisites according examination regulation:			
None			
Recommended prerequisites:			
None			

Objectives:

After successfully completing the master's thesis, students are able to

- to work on a complex engineering problem from the subject area of the study program within a limited period of time and a possibly given budget according to scientific methods in a qualified and independent manner
- systematically and creatively develop solutions for similar problems
- determine and evaluate the limits of the solution presented
- to prepare the problem definition, its classification in an overall context as well as a presentation and discussion of the problem solution and the results in compliance with the rules for scientific texts (stringency, transparency, etc.) and formal criteria
- follow good scientific practice and apply scientific working methods

After successfully completing the master's thesis, students are able to

- to work on a complex engineering problem from the subject area of the study program within a limited period of time and a possibly given budget according to scientific methods in a qualified and independent manner
- systematically and creatively develop solutions for similar problems
- determine and evaluate the limits of the solution presented
- to prepare the problem definition, its classification in an overall context as well as a presentation and discussion of the problem solution and the results in compliance with the rules for scientific texts (stringency, transparency, etc.) and formal criteria
- follow good scientific practice and apply scientific working methods

The students are able to present the contents of their scientific-technical work as well as the strategy of problem solving and the solution methods in a structured way and to represent them convincingly in a subsequent questioning and discussion according to scientific standards.

Content:

The master's thesis is a graduation thesis in engineering specific to the course of study. The topic of the master's thesis is set, supervised and accompanied in terms of content by a professor from the participating universities. The topic can be worked on in practice, e.g. in a company, or in research at the THI.

- scientific analysis of a complex problem specific to the course of study against the background of the state of the art in science and technology.
- literature research, especially considering current international publications in scientific journals
- development of a creative solution concept appropriate to the context of the problem, taking into account current scientific, technical and operational aspects
- comprehensive evaluation of alternative solution concepts and selection of the best solution concept (technical, economic evaluation)
- implementation of the selected solution concept of the complex problem specific to the course of study
- critical and comprehensive analysis of the obtained results using appropriate engineering methods
- project management (especially time and, if necessary, budget management)
- comprehensible and formally correct presentation and documentation of the solution and results
- good scientific practice and scientific working methods

The master's thesis is a graduation thesis in engineering specific to the course of study. The topic of the master's thesis is set, supervised and accompanied in terms of content by a professor from the participating universities. The topic can be worked on in practice, e.g. in a company, or in research at the THI.

- scientific analysis of a complex problem specific to the course of study against the background of the state of the art in science and technology.
- literature research, especially considering current international publications in scientific journals
- development of a creative solution concept appropriate to the context of the problem, taking into account current scientific, technical and operational aspects

- comprehensive evaluation of alternative solution concepts and selection of the best solution concept (technical, economic evaluation)
- implementation of the selected solution concept of the complex problem specific to the course of study
- critical and comprehensive analysis of the obtained results using appropriate engineering methods
- project management (especially time and, if necessary, budget management)
- comprehensible and formally correct presentation and documentation of the solution and results
- good scientific practice and scientific working methods
- familiarization with the scientific-technical problem of the task for the master thesis
- analysis and evaluation of the relevant scientific preliminary work
- development and evaluation of own approaches to the solution
- implementation of the solution
- scientific presentation of the methodology and the solution in written form
- presentation of methodology and results in a lecture as well as their scientific representation in a subsequent survey with discussion.

Literature:

Will be specified at the beginning

Additional remarks:

Important Notes: Keep your supervisors and primary examiners regularly informed of your progress. In particular, clarify their expectations regarding the content of the thesis. A whole semester is estimated for working on the Master's thesis (30 credit points), whereas only 12 credit points are estimated for working on the Bachelor's thesis. This shows that the requirements for the scope and content of a Master's thesis are much higher than for a Bachelor's thesis. In particular, the scientific character should be emphasized more strongly in a Master's thesis:

- statements should, wherever possible, be placed in the context of relevant technical literature.
- in addition to conventional technical literature, sources from current research (e.g., dissertations and conference papers) should be substantially included.
- the graduate's working methods should be purposeful, methodical, and systematic, and should be explicitly documented in the thesis
- quantitative statements, such as measurements, should be investigated and documented using the tools of mathematical statistics.

Important Notes: Keep your supervisors and primary examiners regularly informed of your progress. In particular, clarify their expectations regarding the content of the thesis. A whole semester is estimated for working on the Master's thesis (30 credit points), whereas only 12 credit points are estimated for working on the Bachelor's thesis. This shows that the requirements for the scope and content of a Master's thesis are much higher than for a Bachelor's thesis. In particular, the scientific character should be emphasized more strongly in a Master's thesis:

- statements should, wherever possible, be placed in the context of relevant technical literature.
- in addition to conventional technical literature, sources from current research (e.g., dissertations and conference papers) should be substantially included.
- the graduate's working methods should be purposeful, methodical, and systematic, and should be explicitly documented in the thesis
- quantitative statements, such as measurements, should be investigated and documented using the tools of mathematical statistics.